STAT 527 Advanced Regression Analysis
Fall 2024

Transformations

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Transformations

1. Transforming the response
   - Variance Stabilizing Transformations
   - The Box-Cox method
2. Transforming the predictors
   - Polynomials
   - Regression splines

Reasons to try transformations

- Nonlinearity
- Heteroscedasticity
- Improved fit
- Incorporate a physical law or some other known relationship

Nonlinearity

Let's examine a data set compiled by Allison and Cichetti (1976), which includes body weight (kg) and brain weight (g) of 62 species of mammals. (This data is named `mammals` in the R package `MASS`.) Our goal is to model brain weight as a function of body weight. First, let's take a look at the ratio (brain weight (g)/ body weight (kg)) just for fun ...

```
                   Species      ratio
33          African elephant  0.8584310
4                        Cow  0.9096774
56                       Pig  0.9375000
58           Brazilian tapir  1.0562500
34             Water opossum  1.1142857
22                     Horse  1.2571977
28                   Giraffe  1.2854442
49           Giant armadillo  1.3500000
45                    Jaguar  1.5700000
36                  Kangaroo  1.6000000
19           Asian elephant   1.8072242
42                     Okapi  1.9600000
29                   Gorilla  1.9613527
```

21	Donkey	2.2394441
59	Tenrec	2.8888889
16	Nine-banded armadillo	3.0857143
44	Sheep	3.1531532
5	Grey wolf	3.2892926
18	N.A. opossum	3.7058824
30	Grey seal	3.8235294
6	Goat	4.1576283
37	Yellow-bellied marmot	4.1975309
48	Desert hedgehog	4.3636364
23	European hedgehog	4.4585987
43	Rabbit	4.8400000
8	Guinea pig	5.2884615
50	Rock hyrax-b	5.8333333
3	Mountain beaver	6.0000000
17	Tree hyrax	6.1500000
12	Arctic ground squirrel	6.1956522
13	African giant pouched rat	6.6000000
7	Roe deer	6.6217127
52	Rat	6.7857143
55	Musk shrew	6.8750000
60	Phalanger	7.0370370
25	Cat	7.7575758
38	Golden hamster	8.3333333
57	Echidna	8.3333333
46	Chimpanzee	8.4355828
41	Slow loris	8.9285714

```
51                    Raccoon  9.1417910
24                Patas monkey 11.5000000
62                     Red fox 11.9008264
27                       Genet 12.4113475
20                Big brown bat 13.0434783
1                    Arctic fox 13.1462334
9                        Verbet 13.8424821
10                   Chinchilla 15.0588235
53             E. American mole 16.0000000
31                Rock hyrax-a 16.4000000
15              Star-nosed mole 16.6666667
47                      Baboon 17.0142180
39                      Mouse 17.3913043
32                      Human 21.2903226
61                  Tree shrew 24.0384615
54                    Mole rat 24.5901639
26                      Galago 25.0000000
40             Little brown bat 25.0000000
35                 Rhesus monkey 26.3235294
14 Lesser short-tailed shrew 28.0000000
2                    Owl monkey 32.2916667
11              Ground squirrel 39.6039604
```

Squirrel World?



Linear Regression Model

In order to find a functional relationship between body weight and brain weight, we could consider a simple linear regression model:

$$\mathrm{E}(brain\ weight) \;=\; \beta_0 + \beta_1\, body\ weight$$

Let's see what happens if we try that ...

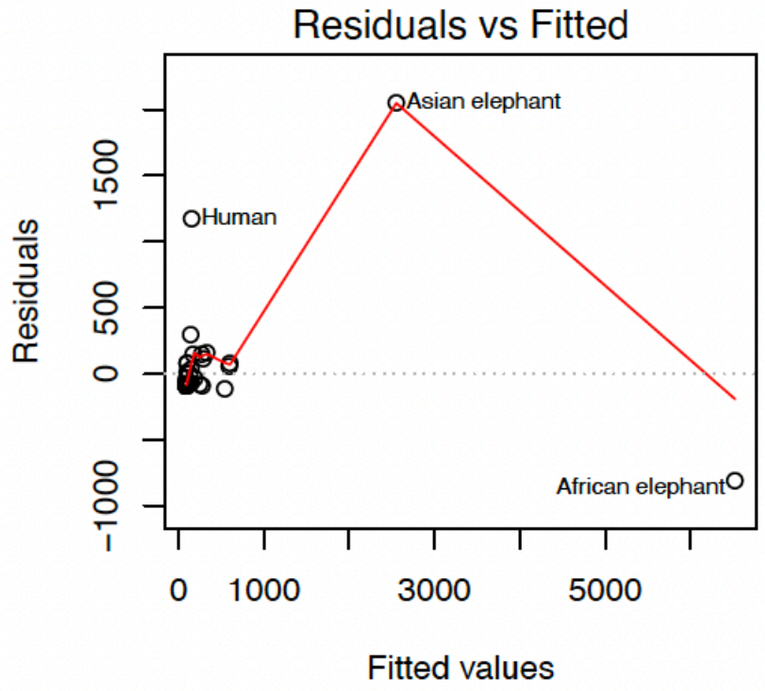
```
> library(MASS)  # contains mammals data set

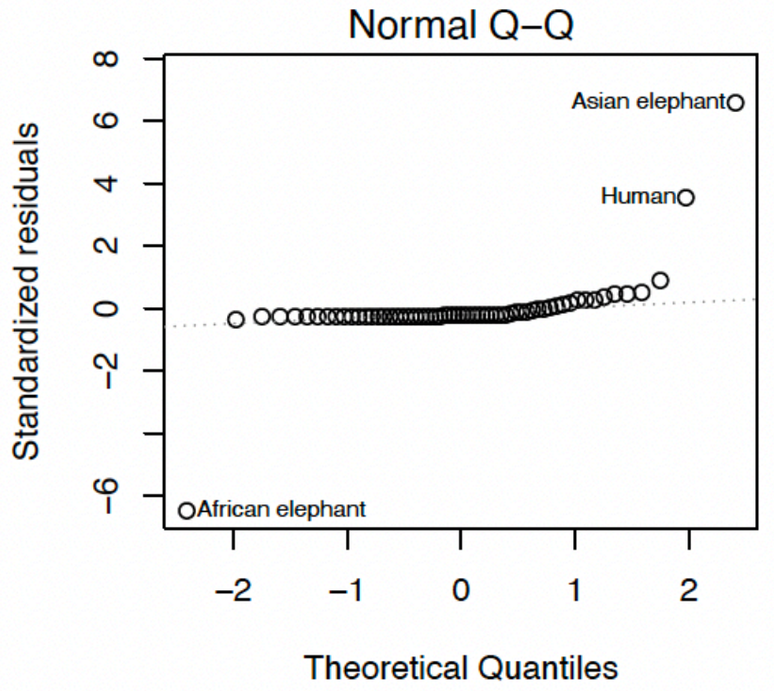
> simplemod <- lm(brain ~ body, data=mammals)
> summary(simplemod)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 91.00440   43.55258    2.09   0.0409 *
body         0.96650    0.04766   20.28   <2e-16 ***
---
Residual standard error: 334.7 on 60 degrees of freedom
Multiple R-squared:  0.8727,    Adjusted R-squared:  0.8705
```

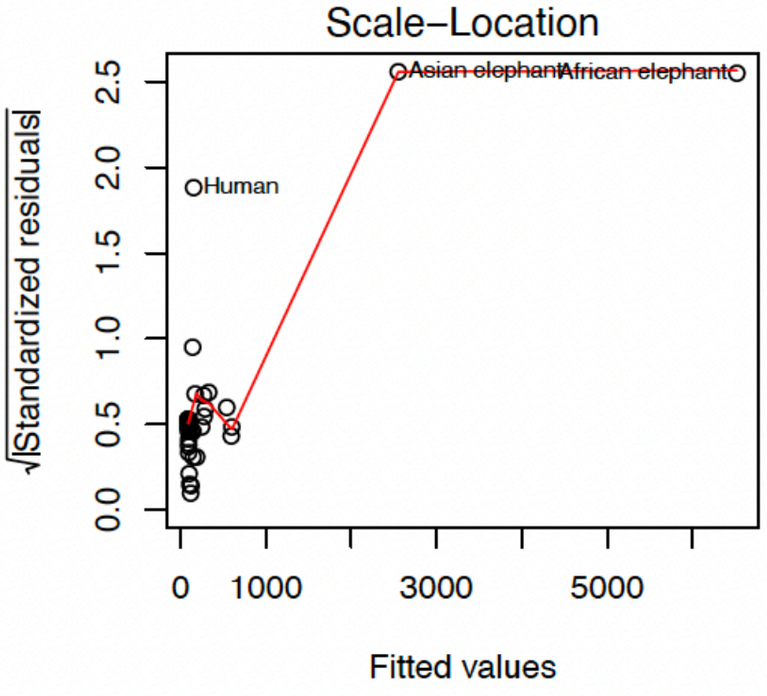
Diagnostics

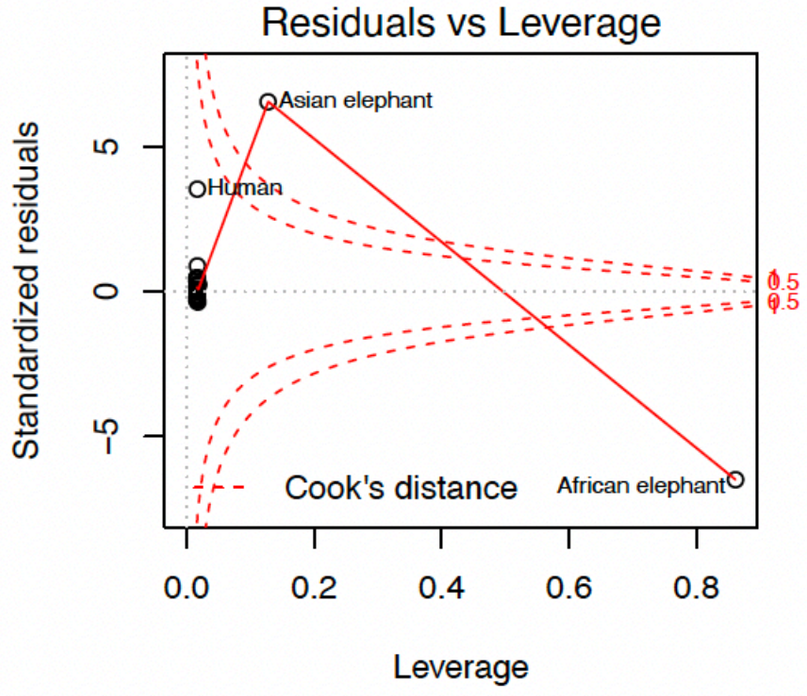
Let us see that the influence of the largest mammals is enormous in the model. Furthermore, there may be possible nonlinearity and heteroscedasticity.

```
> par(mfrow=c(2,2))
> plot(simplemod)
```


Residuals vs Fitted
Residuals
Fitted values
Asian elephant
Human
African elephant
−1000
0
500
1500
0
1000
3000
5000


Normal Q−Q
Standardized residuals
Theoretical Quantiles
Asian elephant
Human
African elephant
−6
−2
0
2
4
6
8
−2
−1
0
1
2

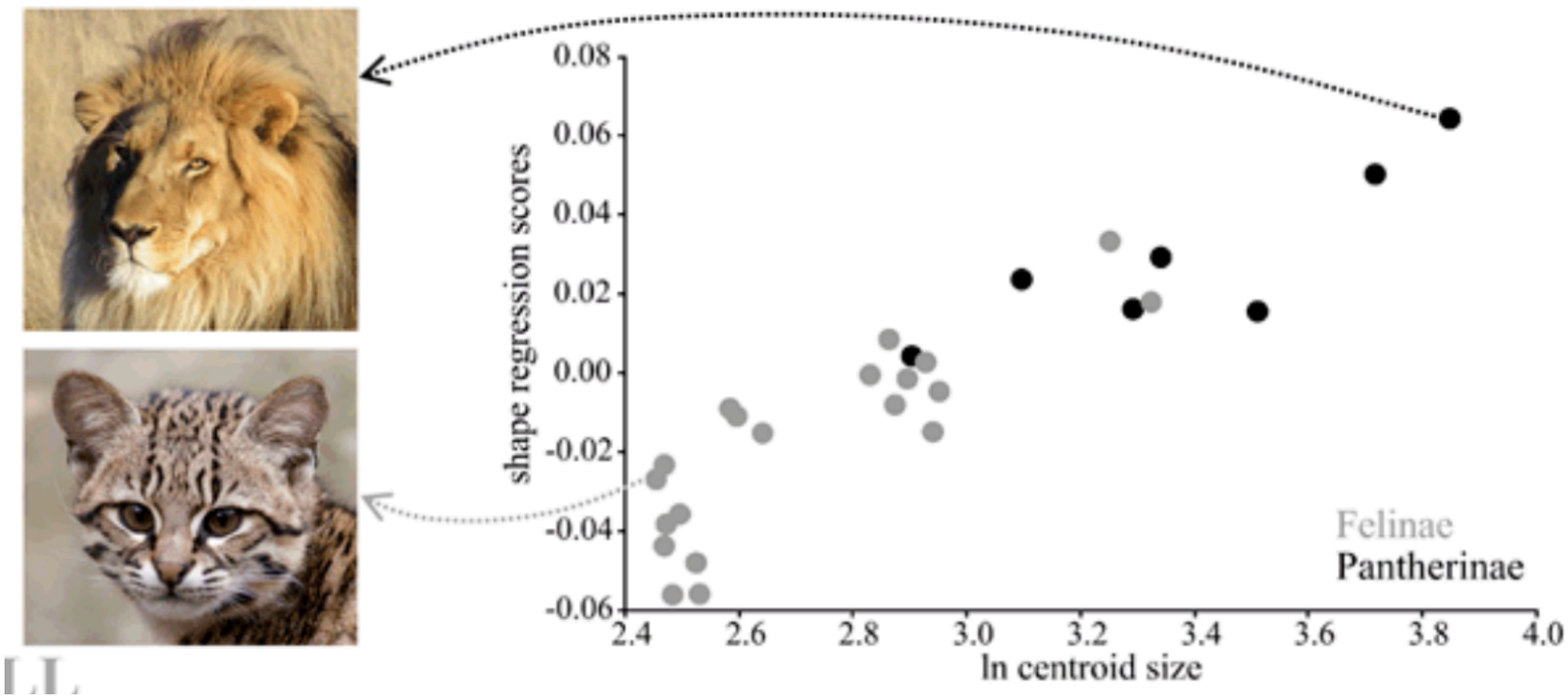

Scale−Location
√|Standardized residuals|
Fitted values
Asian elephant
African elephant
Human
0.0
0.5
1.0
1.5
2.0
2.5
0
1000
3000
5000


Residuals vs Leverage
Standardized residuals
Leverage
Asian elephant
Human
Cook's distance
African elephant
−5
0
5
0.0
0.2
0.4
0.6
0.8

Allometry

- Definition: the growth of body parts at different rates, resulting in a change of body proportions.

- Noted in: Galileo Galilei: Dialogues Concerning Two New Sciences; Julian S. Huxley (1972). Problems of Relative Growth.



Nonlinear Model

Researchers such as Sprent (1972) and Gould (1996) have noted that the following relationship seems to work well for this application,

$$brain\,weight \;=\; \beta_0(body\,weight)^{\beta_1}(\varepsilon)$$

This model asserts that brain weight is proportional to body weight raised to the β_1 power, with a *multiplicative* error ε.

This model is linearizable if we take the logarithm of both sides. That is,

$$\log(brain\,weight) \;=\; \log(\beta_0) + \beta_1 \log(body\,weight) + \log(\varepsilon)$$

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Log transformed model

Let's refit the model with the log-transformed brain and body weights.

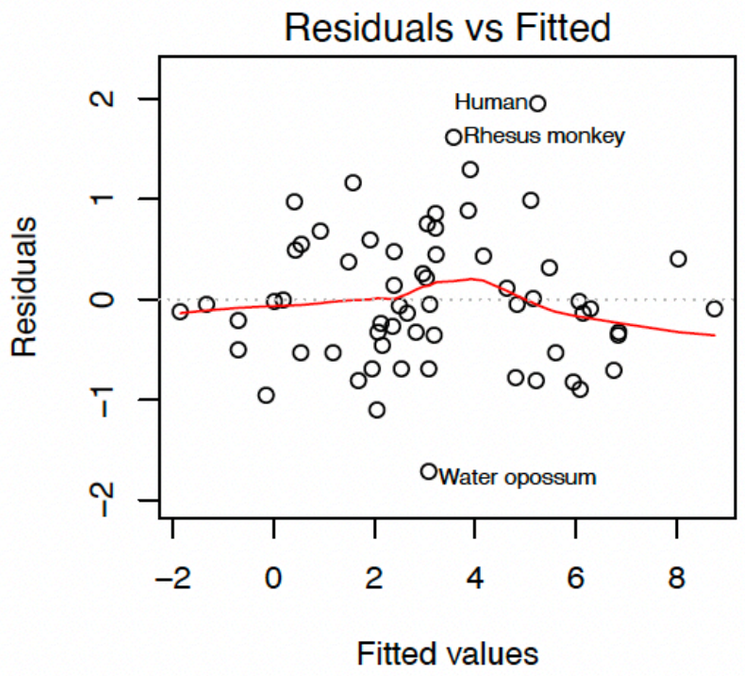
```
> transmod <- lm(log(brain) ~ log(body), data=mammals)
> summary(transmod)

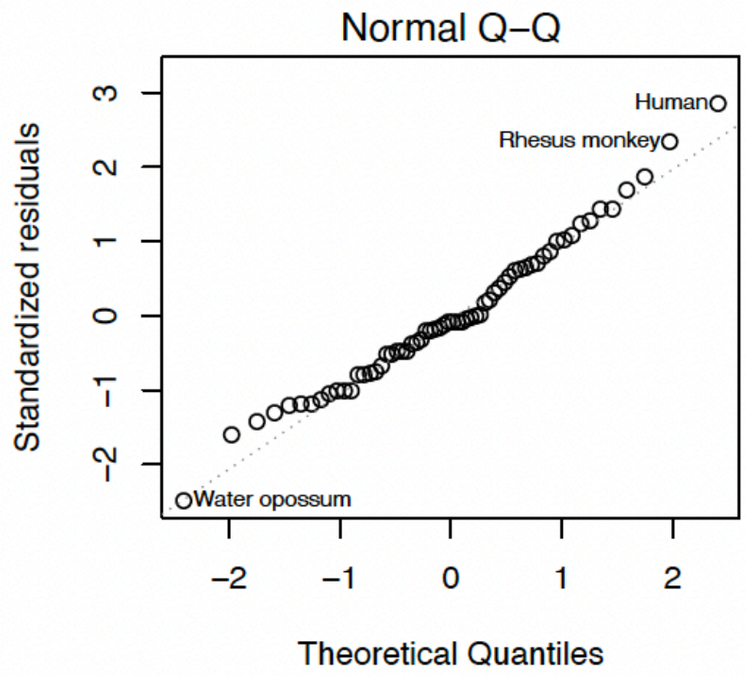
...

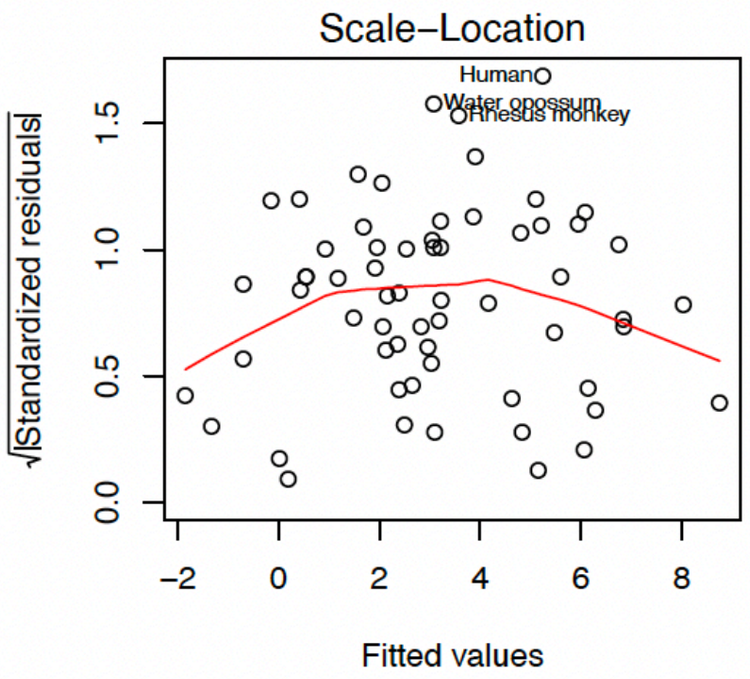
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.13479    0.09604   22.23   <2e-16 ***
log(body)    0.75169    0.02846   26.41   <2e-16 ***

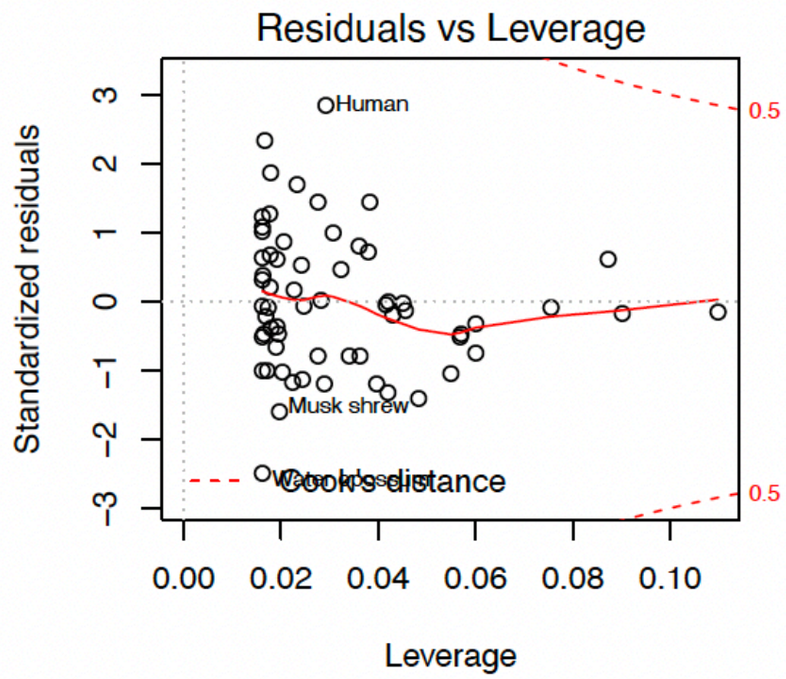
Residual standard error: 0.6943 on 60 degrees of freedom
Multiple R-squared:  0.9208,    Adjusted R-squared:  0.9195


> par(mfrow=c(2,2))
> plot(transmod)
```


Residuals vs Fitted
Residuals
Fitted values
Human
Rhesus monkey
Water opossum


Normal Q−Q
Standardized residuals
Theoretical Quantiles
Human
Rhesus monkey
Water opossum


Scale−Location
√|Standardized residuals|
Fitted values
Human
Water opossum
Rhesus monkey


Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance
Human
Musk shrew
0.5
0.5

Ratio to inspect

The correct ratio to inspect seems to be (brain weight)/(body weight)$^{.7517}$:

	Species	ratio2
34	Water opossum	1.520858
59	Tenrec	2.814293
55	Musk shrew	3.234632
56	Pig	3.458717
58	Brazilian tapir	3.724343
49	Giant armadillo	3.731202
48	Desert hedgehog	3.761669
36	Kangaroo	3.868228
4	Cow	4.180382
23	European hedgehog	4.198504
16	Nine-banded armadillo	4.211607
18	N.A. opossum	4.227777
38	Golden hamster	4.922429
45	Jaguar	4.926059
52	Rat	4.946806
20	Big brown bat	5.112231
8	Guinea pig	5.340215
37	Yellow-bellied marmot	5.940527
22	Horse	5.942845
12	Arctic ground squirrel	6.068698
43	Rabbit	6.076505

```
3              Mountain beaver  6.464175
13 African giant pouched rat  6.600000
39                     Mouse  6.816308
17                Tree hyrax  7.305011
29                   Gorilla  7.372438
14 Lesser short-tailed shrew  7.512981
33           African elephant  7.637951
42                     Okapi  7.720840
60                  Phalanger  7.932543
40           Little brown bat  7.967829
50               Rock hyrax-b  8.017643
5                   Grey wolf  8.026319
21                     Donkey  8.209112
15              Star-nosed mole  8.288266
53            E. American mole  8.410033
44                     Sheep  8.547774
6                       Goat  9.481075
41                Slow loris  9.706565
25                       Cat 10.434533
57                   Echidna 10.946820
30                 Grey seal 11.522299
10                 Chinchilla 12.176431
19            Asian elephant 12.668611
7                   Roe deer 12.934948
51                   Raccoon 13.122608
27                     Genet 13.516680
```

```
61              Tree shrew 13.703643
54                Mole rat 14.584937
31          Rock hyrax-a 15.269386
26                Galago 16.764313
62                Red fox 17.030399
1               Arctic fox 17.794704
9                   Verbet 19.756487
24           Patas monkey 20.370319
11       Ground squirrel 22.413600
46             Chimpanzee 22.518010
2              Owl monkey 26.911795
47                 Baboon 30.541164
35        Rhesus monkey 42.369768
32                 Human 59.324365
```

Variance Stabilizing Transformations

The assumption of constant variance also must be carefully diagnosed: $\mathrm{var}(\varepsilon_i) = \sigma^2$.

In some cases, this assumption does not hold because the variance is approximately a simple function of one of the predictors. For instance: $\mathrm{var}(\varepsilon_i) = \sigma^2 x_{1i}^2$. If there is some reasonable argument for this model, then *weighted* least squares estimation can be used.

More commonly, the constant variance assumption may not hold because the variance is approximately a function of the *mean of* Y, $\mathrm{E}(Y)$. In this case, it may be better to transform Y via a **variance stabilizing transformation**.

Variance Stabilizing Transformations

Transform	Situation	Comments
$\sqrt{Y}$	$\mathrm{var}(Y_i) \propto \mathrm{E}(Y_i)$	Poisson counts
$\log(Y)$	$\mathrm{var}(Y_i) \propto \big(\mathrm{E}(Y_i)\big)^2$	range Y broad
$1/Y$	$\mathrm{var}(Y_i) \propto \big(\mathrm{E}(Y_i)\big)^4$	
$\sin^{-1}(\sqrt{Y})$	$\mathrm{var}(Y_i) \propto \mathrm{E}(Y_i)\big(1-\mathrm{E}(Y_i)\big)$	binomial proportions

Derivations

- Suppose that $\text{var}(Y_i) = f(E[Y_i])$ for some function f and we apply a transformation map ϕ.

- Suppose that $\text{var}(Y_i)$ is small and ϕ is smooth. Then by the Taylor expansion,

$$\begin{aligned} \text{var}(\phi(Y_i)) &\approx \text{var}(\phi'(E[Y_i])(Y_i - E[Y_i])) && (1) \\ &\approx |\phi'(E[Y_i])|^2 \text{var}(Y_i). && (2) \end{aligned}$$

- Therefore, to stabilize the variance, ϕ must satisfy the differential equation

$$|\phi'(y)|^2 f(y) = const. \quad (3)$$

Box-Cox Transformation

Box and Cox (1964) suggested a family of transformations designed to reduce non-normality of the errors. In fact, it often reduces nonlinearity and heteroscedasticity as well.

Suppose each $y_i > 0$, and consider the model

$$\boldsymbol{y}^{\lambda} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

with $\mathrm{var}(\boldsymbol{\varepsilon}) = \sigma^2 \boldsymbol{I}$, where $\boldsymbol{y}^{\lambda}$ represents the *elementwise* power of λ.

(Note: $\lambda = 1$ gives the usual model.)

Choice of λ

The method involves choosing the λ that maximizes the likelihood of the data, assuming normally distributed errors. The method tends to work well for $\lambda \in (-2, 2)$.

By convention, we set $y^\lambda = \log(y)$ for $\lambda = 0$. This is justified because

$$\lim_{\lambda \to 0} \frac{y^\lambda - 1}{\lambda} = \log(y).$$

Let RSS_λ be the residual sum of squares when using y^λ as the response.

Choice of λ - Maximum Likelihood

The (profile) log-likelihood function for $\lambda \neq 0$ is

$$L(\lambda) = n \log(\mid \lambda \mid) - \frac{n}{2} \log(RSS_\lambda) + (\lambda - 1) \sum_{i=1}^{n} \log(y_i)$$

and for $\lambda = 0$,

$$L(0) = -\frac{n}{2} \log(RSS_0) - \sum_{i=1}^{n} \log(y_i).$$

The value that maximizes $L(\lambda)$ is the estimator $\hat{\lambda}$. Box and Cox suggest graphing $L(\lambda)$ versus λ and reading the maximum off the graph. It is customary to round $\hat{\lambda}$ to a simpler nearby value like -1, -1/2, 0, 1/2, or 1.

Derivation of the likelihood

- Suppose that $z = y^{\lambda}$.
- Then $f_Y(y) = f_Z(z)[\lambda y^{\lambda-1}]$.

Savings Data

Let's examine the `savings` data, to see if a power transformation is indicated. First we'll do things the hard way, then the easy way ...

```
> library(faraway)
```

```
> mod <- lm(sr ~ pop15 + pop75 + dpi + ddpi, data=savings)

> # Consider lambda in the range of -2 to 2, and initialize a vector
> # for possible lambda values and a vector to record the log-likelihood

> lambda <- seq(-2, 2, length=400)  # grid of 400 lambda values from -2 to 2
> loglike <- numeric(length(lambda))
> n <- nobs(mod)
```

```
> # Now loop over the 400 values of lambda and record the log-likelihood

> for(i in 1:400){
+   if(lambda[i] != 0){
+     resids <- residuals(lm(sr^lambda[i] ~ pop15 + pop75 + dpi + ddpi,
+                            data=savings))
+     loglike[i] <- n*log(abs(lambda[i])) - (n/2)*log(sum(resids^2)) +
+                   (lambda[i]-1)*sum(log(savings$sr))
+   } else {
+     resids <- residuals(lm(log(sr) ~ pop15 + pop75 + dpi + ddpi,
+                            data=savings))
+     loglike[i] <- - (n/2)*log(sum(resids^2)) - sum(log(savings$sr))
+   }
+ }

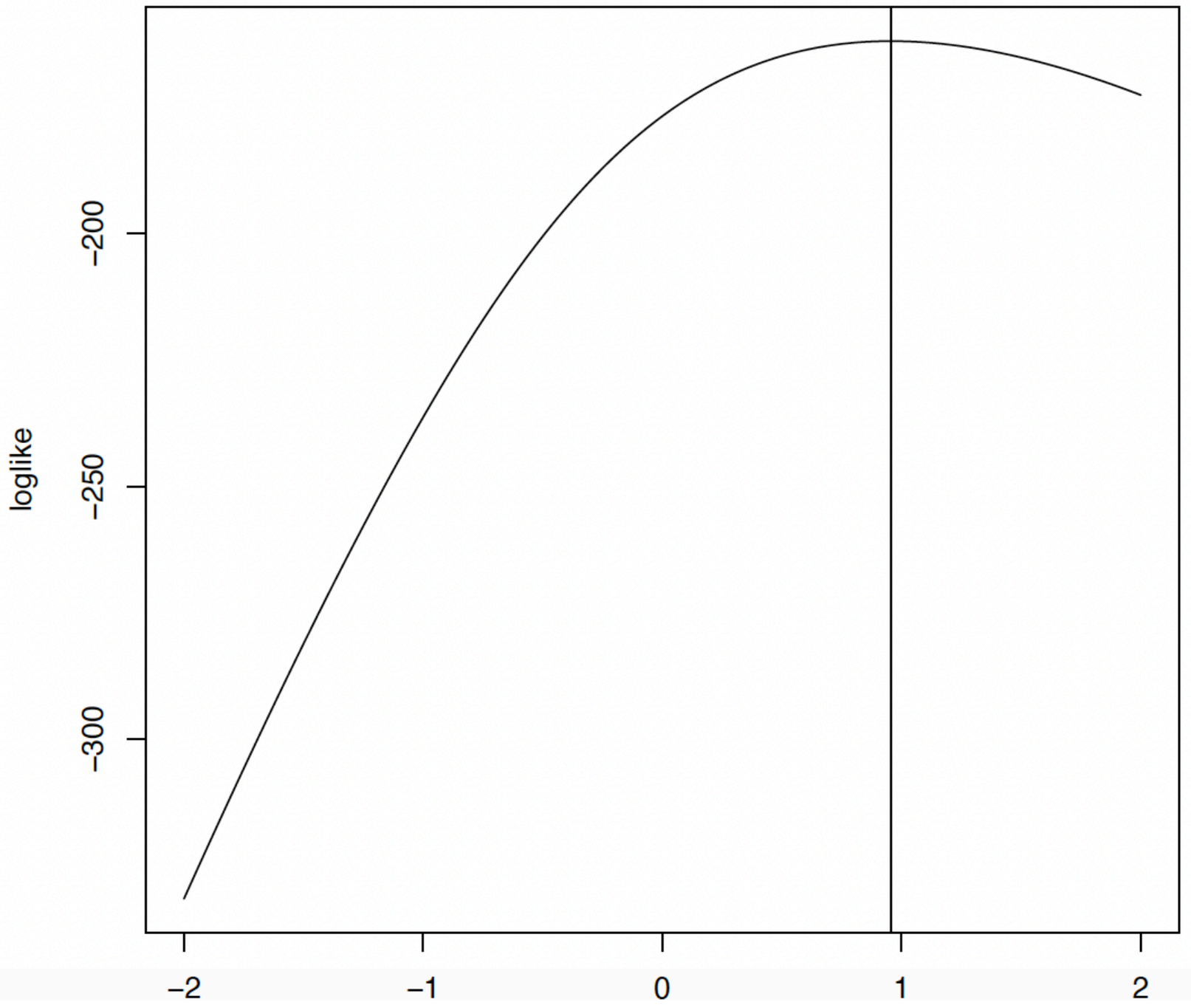
> # Now select lambdahat that maximizes log likelihood

> lambdahat <- lambda[which.max(loglike)]
> lambdahat
[1] 0.9573935

> # Plot loglike versus lambda

> plot(lambda, loglike, type="l")
> abline(v=lambdahat)
```

Log Likelihood vs λ



Choice of λ

No power transformation appears to be needed, since λ is very close to 1.

Now the easy way, using a special function in the `MASS` package ...

```
> library(MASS)  # provides function boxcox

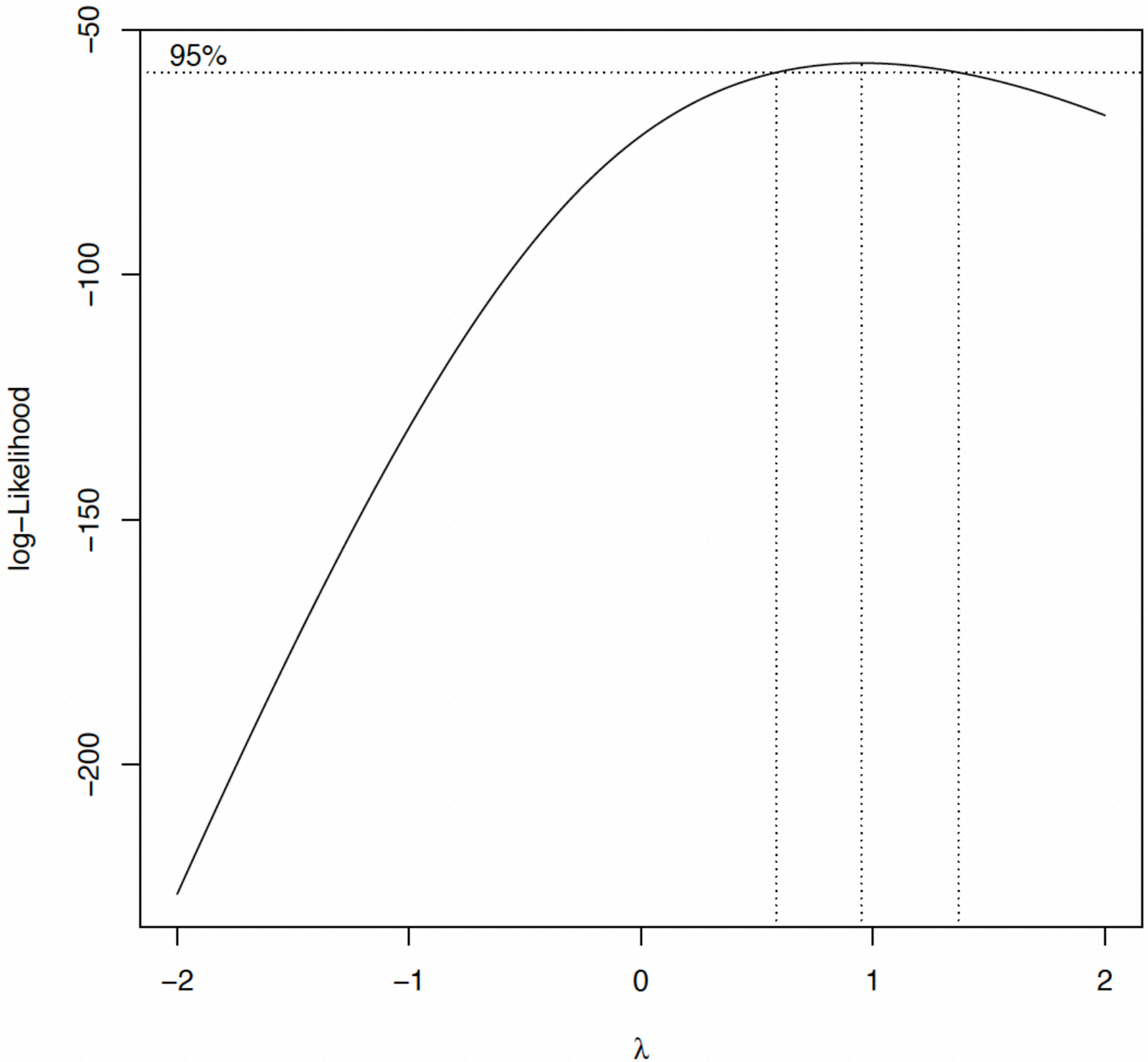
> boxcox(mod)
```

This will produce a plot that also includes an approximate 95% confidence interval for λ ...

Remark on the Confidence Interval

- The interval is asymptotically constructed using Wilk's theorem on the profile likelihood.
- Find all λ s.t.

$$\ln(L(\lambda)) \geq \ln(L(\hat{\lambda})) - \frac{1}{2}\chi^2_{1,1-\alpha}.$$

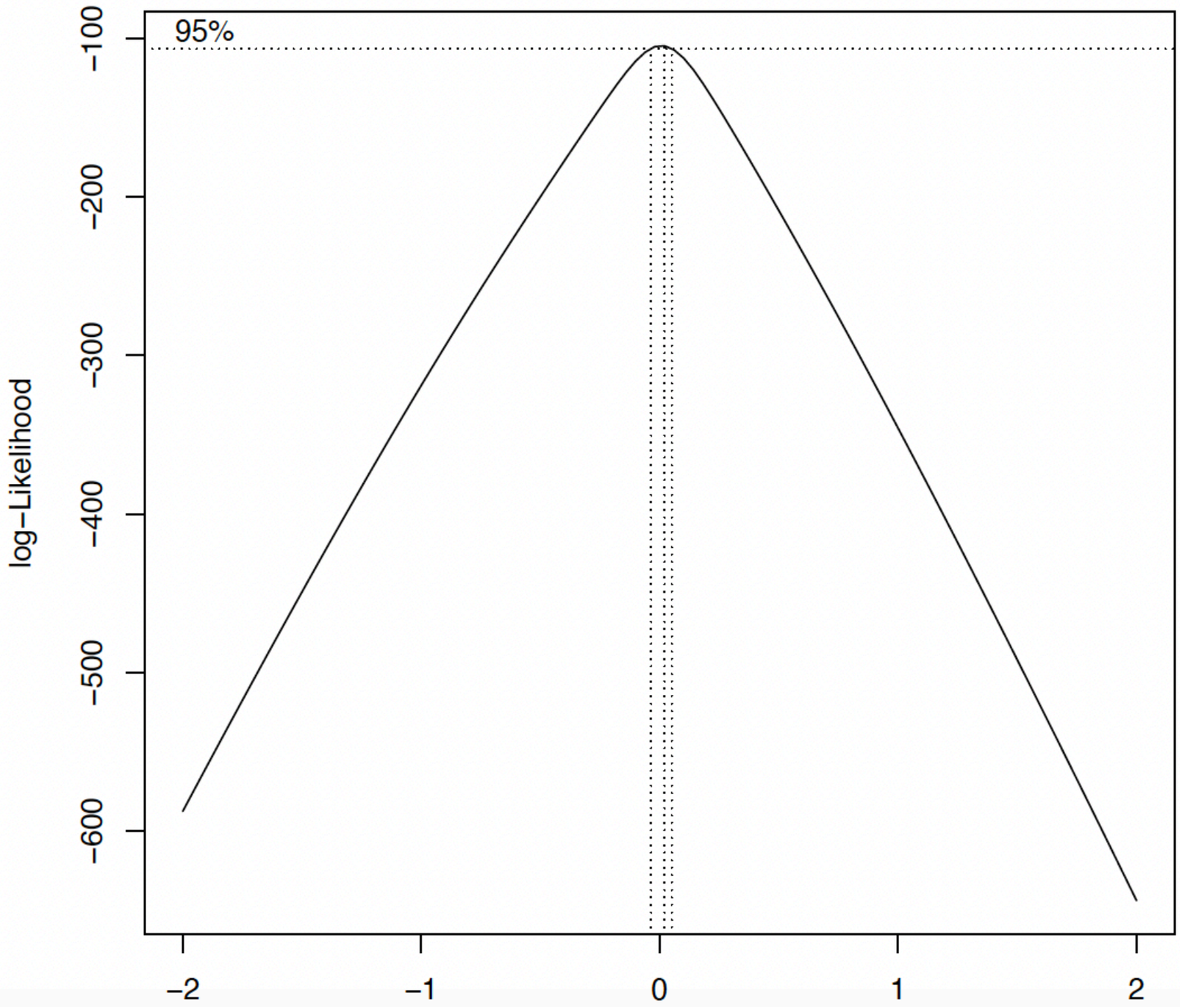

95%
log−Likelihood
−50
−100
−150
−200
−2
−1
0
1
2
λ

Brain and Body weight data

Let's try another example using the brain weight versus body weight data for mammals, using a power transformation for brain weight, after already taking a log-transformation for body weight.

```
> library(MASS)
> regmodel <- lm(brain ~ log(body), data=mammals)
> boxcox(regmodel)
```

Log-likelihood vs λ plot



Thank you!

We gratefully acknowledge the previous course instructor Naveen Narisetty and various online sources for the images and slides presented.